

**CSE 475, Mid Exam # 1, Time: 30 min, Marks: 20 [=10x2]**

1. When do you apply reinforcement learning (RL)? Show with a diagram how RL works.
2. What steps are required to build a supervised ML system?
3. How do you select a best performing ML model?
4. How do you know that your model performs well on unseen data?
5. How is the perceptron related with a biological neuron?
6. What cost function is applied for Adaline?
7. Define **Linearly separable** and **Not linearly separable** problem. Show with a figure.
8. What is the role of learning rate when minimizing cost function with gradient descent? Show with a figure.
9. What is the importance of feature scaling for optimal performance? Give example of feature scaling technique.
10. What is the disadvantage of batch gradient descent?